

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037
<https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Software Engineer with **18 years** delivering end-to-end services for billions of daily interactions across video, e-commerce, and search platforms. Owned the live-streaming recommendation pipeline at YouTube, the real-time auction engine at Whatnot, and the ad-ranking microservice suite at Google, each built on **Go**, **C++**, **Java**, **Kubernetes**, and **Linux**. Reduced latency for ad-ranking by 27% while supporting 12 M QPS, collaborating with product, data science, and security partners. Served on-call for critical incidents, driving kernel-level profiling and memory-footprint reductions that saved \$1.2 M in infrastructure costs annually. Next focus is scaling health-data backends to support millions of wearable users.

Core Skills

Programming Languages: C++, Java, Go, Python, C#, TypeScript, JavaScript, HTML, CSS

Systems & Platforms: Linux, Kubernetes, Docker, gRPC, REST

Cloud & Container: Spinnaker, Memcached, Redis, Prometheus, Grafana

Data & Observability: SQL, BigQuery, Hadoop, JMeter, TeamCity, AWS

Work Experience

Education

B.S. Honors Computer Science

Virginia Tech | 2004 - 2007

Staff Software Engineer, YouTube (02/2025 - Present)

- Designed a distributed live-stream recommendation service handling 150 M daily requests, replacing a monolithic pipeline and cutting average response time from 120 ms to 78 ms, a 35% improvement using **Go**, **Kubernetes**, **Redis**.
- Implemented kernel-level profiling tools that identified memory leaks across 12 services, enabling a 22% reduction in RAM usage and saving \$800 K in cloud spend per year with **C++** and **Linux**.
- Led on-call rotation for the ad-ranking platform, resolving 98% of incidents within SLA and introducing automated rollback scripts that cut mean-time-to-recovery by 40% using **Java** and **Docker**.
- Migrated legacy Python data pipelines to Go microservices, increasing throughput by 48% while maintaining data fidelity across 5 TB daily ingest with **Python** and **Go**.
- Co-authored a real-time monitoring dashboard using **Prometheus** and **Grafana**, giving product owners visibility into latency spikes and reducing post-release bugs by 30%.
- Partnered with security team to integrate token-based authentication, hardening API surface and achieving compliance with SOC 2 within three months using **OAuth** and **TLS**.

Staff Software Engineer, Whatnot (05/2024 - 02/2025)

- Re-engineered the real-time auction engine to run on a Go-based event-driven architecture, supporting 2× peak concurrency and lowering end-to-end latency from 250 ms to 140 ms with **Go**, **Kafka**, **Docker**.
- Built a high-throughput order matching service using **C++** and lock-free queues, processing 1.2 M transactions per second with sub-millisecond latency on **Linux**.
- Introduced automated canary deployments with **Spinnaker** and **Kubernetes**, reducing release failures by 27% and enabling safe rollouts to 1 M active users.

- Created a metrics-driven alerting system in **Prometheus** that cut mean-time-to-detect incidents from 45 min to 12 min, visualized via **Grafana**.
- Mentored a team of 6 engineers on performance testing, leading to a 15% improvement in load-test scores across core services using **JMeter** and **Go**.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024)

- Directed a cross-functional squad to redesign the ad-ranking pipeline, delivering a 27% latency reduction and handling 12 M QPS across 30 countries with **Java**, **Scala**, **Kubernetes**.
- Implemented a distributed cache layer with **Memcached**, decreasing database read load by 40% and saving \$2 M annually in compute costs on **Linux**.
- Established on-call best practices and runbooks, achieving 99.9% uptime for critical services over 3 years using **Python** and **Shell** scripting.
- Integrated an A/B testing framework that enabled rapid feature validation, increasing conversion rate lift by 5% per release with **Go** and **SQL**.
- Embedded statistical scoring service, improving ad relevance and boosting revenue by \$3 M quarterly using **Java** and **BigQuery**.

Software Engineer, Google (02/2011 - 02/2015)

- Developed core search indexing components in **C++**, improving indexing speed by 18% and reducing storage overhead by 12% on **Linux**.
- Created a fault-tolerant logging pipeline using **Java** and **Hadoop**, processing 500 TB of logs per month with zero data loss.
- Optimized network I/O paths, cutting request latency by 22% for high-traffic services with **TCP** and **Linux**.
- Collaborated with UI teams to expose **REST** APIs, increasing third-party integration usage by 30% using **JSON**.

Software Design Engineer II, Microsoft (10/2007 - 12/2011)

- Designed and shipped a cross-platform graphics library in **C++**, adopted by 5 product lines and reducing development time by 25%.
- Implemented performance benchmarks that uncovered 15% CPU waste, leading to code refactor and energy savings across devices using **Python**.
- Built automated build pipelines with **MSBuild**, cutting build times from 45 min to 20 min using **PowerShell**.
- Partnered with QA to introduce continuous integration, decreasing release defects by 33% with **TeamCity** and **Git**.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006)

- Implemented a prototype file-sync service in **C#**, demonstrating 2× speed over legacy tool.
- Wrote unit tests covering 85% of code, improving reliability for subsequent releases using **UnitTest**.
- Presented findings to senior engineers, influencing adoption of the prototype in the main product with **Presentation**.
- Documented API specifications, reducing onboarding time for new interns by 40% using **Markdown**.